

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 19 Taking the Measure of Microbial Systems

19.1 Multiple Choice Questions

- 1) The science of microbial ecology deals with
- A) how microbial communities interact with each other and their environment.
 - B) microbial activity and biodiversity.
 - C) microbial gene regulation mechanisms in response to environmental change.
 - D) how the activity and biodiversity of microbial communities affect microbial interactions with each other and the environment.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

- 2) The methodology of microbial ecology includes
- A) enrichment and isolation of specific microbes.
 - B) cell-staining methods.
 - C) gene, transcript, and protein characterization.
 - D) enrichment and isolation of specific microbes, cell-staining, and characterization of genes, transcripts, and proteins.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.8

- 3) The term used when an organism is studied in its natural environment is
- A) *au naturale*.
 - B) *in vitro*.
 - C) *in situ*.
 - D) *in toto*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.5

- 4) Winogradsky columns are used primarily for enrichment of
- A) aerobic cultures, although occasionally anaerobes do appear.
 - B) anaerobic cultures, although occasionally aerobes do appear.
 - C) aerobes, anaerobes, and phototrophs.
 - D) phototrophs, although occasionally heterotrophs do appear.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

- 5) In an agar dilution tube, microbial colonies grow
- A) at the bottom of the tube.
 - B) on the sides of the tube, although some may be at the surface.
 - C) embedded in the agar, rather than growing on the surface.
 - D) on the surface, but some creep down the sides away from the surface.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

- 6) Which fluorescent molecule enables visualization of living microorganisms?

- A) acridine orange
- B) DAPI
- C) GFP
- D) SYBR® Green

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

- 7) The term axenic is used to describe a(n)

- A) anaerobic culture.
- B) anoxygenic culture.
- C) nutritionally deficient culture.
- D) pure culture.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

- 8) Laser tweezers are most directly used for

- A) enriching a culture.
- B) isolating individual cells.
- C) increasing magnification.
- D) enhancing the growth of fastidious organisms.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.3

- 9) Which approach would help to identify biologically produced sulfur in a sediment sample?

- A) enrichment
- B) isotopic fractionation
- C) microautoradiography
- D) microelectrodes

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

10) Which of the following methods do NOT involve nucleic acid hybridization of a probe to a specimen?

- A) RACE-FISH
- B) FISH-MAR
- C) PhyloChips
- D) T-RFLP

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

11) Which method requires the addition of at least one radioactive chemical?

- A) CARD-FISH
- B) MAR
- C) GeoChips
- D) SIP

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.11

12) Staining methods are suitable for obtaining

- A) information about physiology from natural samples.
- B) quantitative information from natural samples.
- C) phylogenetic information from natural samples.
- D) neither qualitative nor quantitative information from natural samples.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

13) The biggest limitation of traditional light microscopy and electron microscopy is that they are unable to reveal

- A) absolute cell concentration.
- B) cellular components of small microbes because of the limits of magnification.
- C) functional diversity of microorganisms in the habitat under study.
- D) physical microbial interactions as found in nature.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

14) Phylogenetic stains, such as those used in phylogenetic FISH, hybridize with

- A) ribosomal RNA.
- B) mitochondrial DNA.
- C) nuclear DNA.
- D) RNA polymerase.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.5

15) For microbial biodiversity studies, it is common to identify the _____ rather than the _____ as a measure of biodiversity.

- A) organisms themselves / genes
- B) genes / organisms themselves
- C) cell types / genes
- D) cell types / organisms themselves

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

16) Amplification of chromosomal DNA from a single-cell is accomplished by

- A) using PhyloChips.
- B) using multiple displacement amplification.
- C) creating complementary DNA from transcripts of a single-cell.
- D) cloning fragments of the chromosome from the single-cell into a multi-copy plasmid.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.12

17) Well-designed microbial activity measurements can reveal _____ of major metabolic reactions in a habitat.

- A) types
- B) rates
- C) both types and rates
- D) either types or rates (but not both)

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.9

18) Radioisotopes are useful when

- A) very high sensitivity is required.
- B) turnover rates need to be determined.
- C) the fate of portions of particular molecules need to be followed.
- D) very high sensitivity is required, turnover rates need to be determined, and the fate of portions of particular molecules need to be followed.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.9

19) What technique(s) can be used to characterize the phylogenetic composition of a microbial community without culturing any of the members?

- A) green fluorescent protein tagging
- B) stable isotope techniques
- C) radioisotope experiments
- D) fluorescent *in situ* hybridization (FISH)

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.5

20) When dealing with stable isotopes of carbon and sulfur, enzymes tend to favor

- A) the lighter isotope.
- B) the heavier isotope.
- C) the isotopes in about equal proportions.
- D) either the lighter or the heavier isotope, depending on environmental conditions.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

21) Analyses of sulfur isotopes have been used as evidence against life on the Moon, because the sulfides in lunar rocks have

- A) an isotope composition most similar to marine sulfate, which is enriched in ^{34}S .
- B) an isotope composition most similar to marine mud, which is enriched in ^{34}S .
- C) an isotope composition most similar to igneous rocks.
- D) exactly equal amounts of ^{34}S and ^{32}S .

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 19.10

22) Which technique would be used to estimate the concentration of naturally occurring *Escherichia coli* in a wastewater sample?

- A) agar dilution tube method
- B) laser tweezers method
- C) most probable number (MPN) method
- D) cell tagging using GFP method

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

23) Which molecular method allows for the detection of a specific organism within a diverse microbial community?

- A) fluorescent DNA staining using DAPI
- B) viability staining
- C) enrichment techniques
- D) fluorescent *in situ* hybridization (FISH)

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.5

24) For what purpose would a microbial ecologist use a microsensor?

- A) to determine the concentration of cells in small increments in a sediment or water column
- B) to determine the expression of a particular gene in a microbial community
- C) to quantify the uptake of a radioactive material
- D) to determine the concentration of O₂, N₂O, or other chemical species in small increments in a sediment or water column

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.9

25) Microarrays are useful for assessing

- A) gene expression only.
- B) enzyme activity.
- C) the presence of specific rRNA sequences only.
- D) gene expression and the presence of specific rRNA sequences.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.7

26) Genes encoding _____ are widely used as target genes in molecular biodiversity studies.

- A) ATPases
- B) DNA polymerase III
- C) 16S rRNA
- D) RNA polymerase

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

27) How do viability stains usually distinguish between living cells and dead cells?

- A) The chemical fluoresces only when bound to ribosomes that are actively synthesizing proteins.
- B) The chemical stain is taken up by living cells and once metabolized becomes fluorescent.
- C) The dye specifically targets intact cytoplasmic membranes.
- D) The dye's fluorescence is quenched by dead cells.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

- 28) Why is it necessary to have a standard for the interpretation of stable isotope ratios?
- A) A standard is necessary to calculate how rapidly the stable isotopes undergo radioactive decay.
 - B) A standard is necessary to determine the stable isotope ratio in the absence of biological activity.
 - C) A standard is necessary because samples are only presented in comparison with other samples of the same age.
 - D) A standard is necessary to determine the age of the sample.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

- 29) What unit is isotopic fractionation reported as?

- A) parts per billion
- B) parts per hundred
- C) parts per thousand
- D) parts per trillion

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

- 30) Pure cultures can be obtained from an enrichment culture using

- A) streak plates only.
- B) the agar dilution method only.
- C) liquid dilution methods only.
- D) streak plates, the agar dilution method, or liquid dilution methods.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

- 31) Metatranscriptomics and metaproteomics are favored over metagenomics when _____ is of more interest than _____.

- A) gene presence / phylogeny
- B) phylogeny / gene presence
- C) function or gene expression / gene presence
- D) gene presence / function or gene expression

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.8

32) What are activity measurements?

- A) Activity measurements are any measurements taken *in vitro*.
- B) Activity measurements are any measurements taken *in situ*.
- C) Activity measurements are collective estimates of the physiological reactions in an entire microbial community.
- D) Activity measurements are an example of metagenomics, which can be used when genetic analysis is difficult.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.9

33) Interpreting stable isotopic fractionation depends on the fact that most enzymes preferentially bind _____ isotope substrates over _____ isotope substrates.

- A) nonradioactive / radioactive
- B) light / heavy
- C) heavy / light
- D) radioactive / nonradioactive

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

34) Thirty organisms containing the same phylogenetic marker, with identical or similar orthologous genes, are all considered to be within the same

- A) phylotype.
- B) species.
- C) functional group.
- D) phylogenetic tree.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.6

35) Genomic analysis of *Prochlorococcus* has revealed

- A) the presence of bacteriorhodopsin.
- B) that no regions are shared between cultured representatives and environmental populations.
- C) complete congruence of genes and genome structure within the same phylotype.
- D) high levels of gene conservation between cultured representatives and environmental populations.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.8

36) Robotic systems used to quickly test the response of microbes in microtiter plates to thousands of combinations of nutrients is an example of a(n) _____ technology.

- A) next generation sequencing
- B) stable isotope
- C) genomic
- D) high-throughput

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.3

37) When laser tweezers are used, the laser beam can trap

- A) an isolated single-copy genome in a capillary tube.
- B) a single microbial cell in a capillary tube.
- C) a single microbial colony on a plate.
- D) multiple cells of the same type on a plate.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.3

38) Which of the following techniques can link prokaryotic phylogeny and metabolic function?

- A) DAPI staining
- B) stable isotope probing only
- C) environmental genomics only
- D) environmental genomics and stable isotope probing

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

39) Diversity is often _____ when T-RFLP is used.

- A) underestimated
- B) overestimated
- C) randomly overestimated or underestimated
- D) impossible to estimate

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

40) In microbial ecology studies, the FISH technique identifies organisms based on

- A) the presence of green fluorescent protein.
- B) fluorescence of unique cofactors or pigments under certain wavelengths of UV light.
- C) hybridization of a fluorescent probe to specific ribosomal RNA sequences.
- D) the presence of luciferase.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.5

41) The overall purpose of DGGE is to
A) separate genes that differ in length.
B) measure the activity of cells that use DGGE as an energy source.
C) detect cells that utilize a specific radiolabeled substrate.
D) separate genes of the same length that have different sequences, resulting in different denaturation profiles.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

42) Stable isotope probing (SIP) is a method that
A) causes cells that hybridize with a fluorescent probe to fluoresce under UV light.
B) compares the isotopic ratios of different compounds in the environment to determine whether they are of biological or geological origin.
C) uses substrates containing heavy isotopes to label the DNA of organisms that use those substrates.
D) compares the transcription of specific isotope labeled genes under different conditions.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

43) The element(s) that have proven most useful for stable isotope studies are
A) carbon only.
B) nitrogen only.
C) sulfur only.
D) carbon, sulfur, and sometimes nitrogen.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

44) The $^{13}\text{C}/^{12}\text{C}$ isotopic ratio of geological rocks of different ages has been used to examine whether previous biological activity occurred because compounds of biological origin
A) do NOT contain ^{13}C .
B) have a higher proportion of ^{13}C than compounds of geological origin.
C) have a lower proportion of ^{13}C than compounds of geological origin.
D) are not well preserved in geological samples.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.10

45) Transcription of various genes under particular environmental conditions in natural populations can be studied using

- A) CARD-FISH.
- B) stable isotope analysis.
- C) T-RFLP.
- D) green fluorescent protein (GFP).

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.5

46) You would like to test the hypothesis that ammonia-oxidizing *Archaea* are more active and abundant than ammonia-oxidizing *Bacteria* in coastal marine water. What technique(s) or experiment(s) could you use to test this hypothesis?

- A) environmental proteomics
- B) fluorescent *in situ* hybridization (FISH)
- C) stable isotope probing using ^{15}N labeled ammonia
- D) stable isotope ratios of ammonia and nitrite

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.10

47) Which of the following hypotheses could be tested using microautoradiography fluorescence *in situ* hybridization (MAR-FISH)?

- A) Bacteriorhodopsin uses light energy to translocate protons.
- B) *Archaea* are the main autotrophs in marine water samples taken 300 m below the surface of the ocean.
- C) The microbial community of agricultural soils is more phylogenetically diverse than the microbial community of native prairie soils.
- D) The ammonia monooxygenase gene is present in members of the phylum *Verrucomicrobium*.

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 19.11

48) How would you label and separate the DNA of unknown methylotrophs in a sample?

- A) Incubate the sample with ^{13}C -methane.
- B) Hybridize the DNA to FISH probes.
- C) Use DGGE to separate the strands of DNA in the sample.
- D) Use T-RFLP to identify the methylotroph DNA in the sample.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.11

49) Single-cell genomics may help us

- A) predict which microbes will be abundant based on the environmental conditions.
- B) understand the physiology of uncultured microbes.
- C) develop better enrichment culture methods.
- D) understand the physiology of uncultured microbes and develop better enrichment culture methods.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.12

50) Culture-independent approaches avoid issues associated with

- A) detecting very rare phylotypes.
- B) conflating phylogenetic and functional diversity.
- C) growing organisms in the lab.
- D) next generation sequencing.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

51) Which of the following statements is FALSE concerning enrichment bias?

- A) Dilution of the inoculum may help to avoid enrichment bias in some cases.
- B) Enrichment bias is impossible to overcome in some oligotrophic environments.
- C) Enrichment bias resulted in the underestimation of prokaryotic diversity.
- D) Enrichment bias favors organisms that grow quickly.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 18.1

19.2 True/False Questions

1) One way to study a microbial ecosystem is to isolate microorganisms from it and study their properties in laboratory culture.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

2) Strict anaerobic organisms are not suited for GFP reporters due to the protein requiring oxygen to fluoresce.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

3) The purpose of the enrichment culture technique is to enrich for organisms with specific physiological properties from a microbial community.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

4) Enrichment cultures can be established by placing the inoculum into selective media and incubating under specific conditions.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

5) If properly constructed, a Winogradsky column can be used to isolate pure cultures.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

6) Once a putative pure culture has been obtained from the environment, it is no longer necessary to check its purity.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

7) Organisms in natural environments can be detected by assaying for the presence of their genes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.8

8) Traditional enrichment culturing circumvents issues associated with "weed" species from overcoming the population.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

9) The goal of environmental genomics is to detect as many genes as possible and then, if possible, to determine to the phylogeny of the organisms to which the genes belong.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

10) FISH technology can employ only a single phylogenetic probe.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.5

11) The MPN technique is rarely applied in the food industry because any microbial presence in food is unsafe.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

12) The Winogradsky column is a method for assessing the entire gene complement of a habitat, revealing both the biodiversity and metabolic capabilities of the microbial community at the same time.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

13) In natural samples it is often difficult to differentiate live cells from dead cells or cells from nonliving matter without special dyes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

14) DGGE and T-RFLP are both molecular microbial ecology methods that use polymerase chain reaction (PCR) to amplify specific genes to assess diversity.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

15) By using simple microscopy techniques, it is possible to determine if a pure culture is contaminated with another organism.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.2

16) Fluorogenic dyes, autofluorescence, and genetically engineered autofluorescence can be used to identify cells.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.4

17) The diversity of microorganisms in culture collections is representative of the diversity of microorganisms found in nature.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.1

18) Metaproteomics is a more developed field than metatranscriptomics due to the high sensitivity of mass spectrometry.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.8

19) Phototrophic organisms in SIP experiments are often observed releasing $^{14}\text{CO}_2$ as a way of measuring photosynthetic activity.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.9

20) Combined with autofluorescence or fluorescent DNA probes, flow cytometry can separate or count populations of cells from aquatic environments.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.12

19.3 Essay Questions

1) What are the pros and cons of traditional enrichment culture techniques? What advances in the last decade are improving enrichment culture techniques to aid the cultivation of uncultured microorganisms?

Answer: The pros of traditional enrichment culture techniques are that enrichment culture can select for organisms with very specific physiologies for generating energy and/or for tolerating specific growth conditions. For instance, someone interested in anoxic photosynthesis in arctic environments would incubate samples under specific wavelengths of light in a minimal media at a very low temperature. The cons of enrichment culture are that fast-growing microbes ("weeds") often out-compete other microbes in the sample, thus underestimating diversity. In addition, it is often the slow growing microbes that are most prevalent in the environment and those organisms may be more difficult to culture through traditional enrichment techniques. Another con of traditional enrichment culture is that laboratory conditions may actually kill the most prevalent microbes (for instance, if growth conditions are not correct) or we may not be supplying the correct carbon, energy, nutrient, or cofactors for growth of the most prevalent microbes.

The first major change in the last decade is our ability to detect and sequence microorganisms in environments with cultivation-independent techniques. This has allowed us to avoid the bias of enrichment cultures and more accurately assess and catalogue the microorganisms present in a sample. Metagenomic techniques have been particularly helpful because the genetic content of a population can be analyzed to gain clues as to what growth conditions, nutrients, cofactors, and energy sources a microorganism may require for growth. Other advances that have improved enrichment culture techniques include high-throughput screening in 96 well plates that allows us to test hundreds of combinations of growth substrates, nutrients, and cofactors. Methods of diluting or separating single cells or populations of similar cells into test plates will also aid in cultivation of the specific microorganisms we would like to study. Lastly, using nucleic acid based techniques (FISH, T-RFLP, etc.), we can monitor enrichment cultures quickly and accurately to determine which growth conditions encourage the growth of the microorganisms we wish to study.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.1, 19.8

2) Describe how FISH can be used to determine which cells within a microbial community are actively degrading toluene. Assume the genes involved in the metabolic pathway have been sequenced from several microorganisms and vary only slightly from one another.

Answer: A conserved gene sequence would need to be identified and made into a FISH probe. It is likely CARD-FISH will need to be used to observe a signal. The community could be grown on toluene and exposed to the probe, where microbes expressing the genes for the toluene degradation pathway would hybridize to the CARD-FISH probe and could be visualized using fluorescence microscopy.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.5

3) We know that there are over 50 major phyla in *Bacteria*; however, only about 25 of these phyla have cultured representatives. Why is this? How are scientists attempting to culture representative from the 25 phyla that do not currently have any organisms in culture?

Answer: We know that more phyla of *Bacteria* exist than what is represented in culture because of culture-independent techniques that rely on nucleic acid to detect and classify microbes.

Scientists are trying to culture these organisms by using metagenomic sequencing and genome sequencing of single cells to find out what genes the uncultured organisms possess (or lack) so that they can design the appropriate culture conditions for the uncultured organisms. High throughput methods that sort single cells using robots, 96 well plates, and plate readers to detect specific organisms are also being used to test hundreds of different growth conditions for one particular cell type in the hopes of finding the right conditions for growth.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 19.1, 19.12

4) How can NanoSIMS be used to determine which *Roseobacters* most strongly use DMSP as a sulfur source within a coastal marine environment? Explain the NanoSIMS methodology in your answer.

Answer: NanoSIMS is a highly sensitive technique that can be combined with other methods to quantify the relative proportions of isotopic atoms such as ^{34}S incorporated into various isolated organisms from ^{34}S -DMSP exposure. The organisms containing the highest ^{34}S are likely those that assimilate DMSP the most. In NanoSIMS, a high-energy ion beam is directed onto a sample to break apart most chemical bonds such that secondary ions released represent the composition of the sample. The various secondary ions are quantified with a mass spectrometer.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.11

5) Using specific examples, when is it appropriate to use CARD-FISH rather than the traditional FISH method and vice-versa?

Answer: CARD-FISH is especially useful when the target nucleotide sequence is at a relatively low concentration such that the fluorescent signal using traditional FISH is not strong enough to detect. While 16S rRNA-targeted probes might work in observing cells with a high copy number, other low copy number target sequences such as mRNAs (e.g., catabolic pathway transcripts) usually require CARD-FISH to detect the target sequence.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.5

6) Propose an experiment to identify members capable of assimilating monomers of lignin within a microbial community.

Answer: SIP could be used, which would require isotopically labeled lignin monomers to feed a microbial community. Using labeled carbon as an example, the heavy (^{13}C) atoms would be incorporated into members that have assimilated the labeled lignin monomers such that their ^{13}C -DNA could be separated from ^{12}C -DNA lacking the carbon isotope. The DNA could then be studied to identify the individual members involved in this metabolic process.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.10

7) A study of the human gut performed in 1982 used various types of plates under different growth conditions to describe and enumerate the microbes in the human gut. They concluded that the majority of the microbes in the human gut are *Gammaproteobacteria*. A study conducted in 2010 using next-generation sequencing to characterize 16S rRNA sequences concluded that the majority of the microbes in the human gut are *Firmicutes* and *Bacteroidetes*. Why did these studies have different conclusions? Which study do you think is most accurate?

Answer: The studies do not agree because of enrichment / culture bias. Many organisms are easy to grow in the lab and grow faster than other organisms on agar plates or liquid media. Some of the most numerous organisms in the environment or the human body do NOT grow well, or they do NOT grow AT ALL in the lab conditions tested so far. This creates a biased view of microbial community composition. The first study suffered from culture bias.

Gammaproteobacteria tend to be very easy to grow in the lab. The second study is probably more accurate, as it is based on sequencing nucleic acid directly and circumvents growth of the organisms in order to detect them.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 19.1

8) Describe what an ITS region is in the context of prokaryotes, and explain the significance of the ITS region for microbial ecology.

Answer: The ITS region is the nucleotides between the 16S and 23S rDNA. Differences in ITS region length among microbes are commonly exploited in the ARISA method to observe community composition. They are also sequenced when 16S rDNA sequencing cannot resolve closely related microorganisms.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 19.6

9) What prior knowledge is required to use a phylochip to study the abundance of the different microbes present within a microbial community? How does this cause bias when using a phylochip to analyze microbial community structure?

Answer: To construct the phylochip, the 16S rRNA nucleotide sequences must be known to detect the microorganisms. Without the 16S rRNA nucleotide sequences, the probes cannot be synthesized. If there are organisms present in the sample that do NOT match the probes on the chip, they will not be detected. This is one bias introduced by a phylochip—novel organisms that have not been sequenced will be completely overlooked.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 19.7

10) Propose a set of procedures or steps to sequence an individual cell's genome present in lake water.

Answer: First, individual cells would need to be sorted through either flow cytometry or use of laser tweezers. To obtain sufficient quantities of DNA for genome sequencing, the individual cell's genome must then be amplified. Traditional PCR will not suffice, so multiple displacement amplification (MDA) is required to randomly replicate the cell's genome using a bacteriophage DNA polymerase.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 19.12